

University of Cape Town
Department of Mathematics and Applied Mathematics

2IA Final Exam
9th November, 2023
Time: 2 hours
Total marks: 105, Full marks: 100

- Read the instructions carefully.
- You have 2hrs to complete the exam.
- Make sure that no pages or questions are missing.
- Show all your work or reasoning. Answers without work or reasoning shown will receive no mark.

Surname: _____

First name: _____

Student number: _____

Signature: _____

Do not write below this line:

Question:	1	2	3	4	Total
Marks:	15	40	26	24	105
Score:					

1. Let G be a nontrivial group (that is, $G \neq \{1\}$).

(a) 10 marks For $g \in G$, prove that $g^2 = 1$ if and only if $g^{-1} = g$.

(b) 5 marks Suppose that $|G|$ is even (notice that G is then finite). Prove that there exists $a \in G$ with $a \neq 1$ such that a has order 2 (that is, $a^2 = 1$).

Solution: (a) Suppose first that $g^2 = g \cdot g = 1$. Then multiplying by the inverse of g on both sides of the equation we get that

$$g^{-1}g^2 = g^{-1} \Rightarrow g = g^{-1}.$$

Conversely, assume that $g^{-1} = g$. Then $g^2 = g \cdot g = g \cdot g^{-1} = 1$, finishing the proof.

(b) Using (a), if g an element in G is such that $g^2 \neq 1$, then $g^{-1} \neq g$. We could pair all these elements, obtaining an even number in total. All of these elements together with the identity produce an even number of elements. Since we are assuming the order of G is even, the group G must contain an element $a \neq \{1\}$ such that $a^2 = 1$.

2. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**

- (a) 5 marks Suppose that m and n are two positive integers. If $d > 1$ satisfies that $d = am + bn$ for some integers a and b , then $d = \gcd(m, n)$.
- (b) 5 marks The set of all permutations S_3 is an abelian group.
- (c) 5 marks Any permutation $\sigma \neq \varepsilon$ can be factorised in a **unique way** as a **product of transpositions**.
- (d) 5 marks The empty set is a group.
- (e) 5 marks $\mathbb{Z}_2$ is a subgroup of $\mathbb{Z}_3$.
- (f) 5 marks If H and K are subgroups of G , then $H \cup K$ is also a subgroup of G .
- (g) 5 marks Every group of order prime is cyclic.
- (h) 5 marks If $K \subseteq H \subseteq G$ are finite groups, then $|G : K| = |G : H| \cdot |H : K|$.

Solution: (a) **False:** The converse of Bézout's Lemma does not hold. Take for example, $m = 4$, $n = 8$. Then $\gcd(4, 8) = 4$ and 12 satisfies that $12 = 1 \cdot 4 + 1 \cdot 8$.

(b) **False:** The transpositions $\alpha = (1 \ 2)$ and $\beta = (1 \ 3)$ are in S_n and they do not commute. In fact:

$$\begin{aligned}\alpha\beta &= (1 \ 2)(1 \ 3) = (1 \ 3 \ 2), \\ \beta\alpha &= (1 \ 3)(1 \ 2) = (1 \ 2 \ 3).\end{aligned}$$

(c) **False:** For instance, we proved in class that the 5-cycle $\alpha = (1 \ 2 \ 3 \ 4 \ 5)$ can be written as

$$\alpha = (1 \ 2)(2 \ 3)(3 \ 4)(4 \ 5) = (1 \ 5)(1 \ 4)(1 \ 3)(1 \ 2).$$

(d) **False:** The empty set is not a group since it does not contain an identity element.

(e) **False:** A subgroup (is by definition) a subset of a group such that it is a group itself with the same operation. In this case, $\mathbb{Z}_2$ is **NOT** a subset of $\mathbb{Z}_3$, and so it does not make sense to think about subgroups.

Be very careful here, we are using the same notation here to denote (for example) $\bar{0} \in \mathbb{Z}_2$ and $\bar{0} \in \mathbb{Z}_3$ but these are not the same! $\bar{0}$ viewed as an element of $\mathbb{Z}_2$ denotes the even numbers, while $\bar{0} \in \mathbb{Z}_3$ denotes all the integers that have a remainder of 0 when divided by 3; that is the multiples of 3 and these two sets are not equal!

(f) **False:** Take, for example, $G = \mathbb{Z}$, $H = 2\mathbb{Z}$ and $K = 3\mathbb{Z}$. Then $5 = 2 + 3 \in H \cup K$ but $5 \notin H$ and $5 \notin K$.

(g) **True:** Suppose that G is a group of order p , where p denotes a prime. Then G contains an element g such that $g \neq 1$. From Lagrange's Theorem, we get that $|\langle g \rangle|$ divides $|G| = p$. So either $|\langle g \rangle| = 1$ or $|\langle g \rangle| = p = |G|$. Notice that $|\langle g \rangle| = 1$ can not happen since this would imply $g = 1$. Thus, $|\langle g \rangle| = p = |G|$, which yields $G = \langle g \rangle$, that is, G is cyclic.

(h) **True:** Since all the groups are finite (from the proof of Lagrange's Theorem) we have that

$$|G : H| \cdot |H : K| = \frac{|G|}{|H|} \cdot \frac{|H|}{|K|} = \frac{|G|}{|K|} = |G : K|.$$

3. (a) 3 marks Complete the missing parts (marked with a $\star$) in the statement of the **Fundamental Theorem of Finite Cyclic Groups**.

FUNDAMENTAL THEOREM OF FINITE CYCLIC GROUPS. Let $G = \langle g \rangle$ be a cyclic group of order n .

- (1) If H is a subgroup of G , then $H = \langle g^d \rangle$ for some $(\star)$.
- (2) If H is any subgroup of G with order $|H| = k$, then $(\star)$.
- (3) If $k|n$, then $\langle g^{n/k} \rangle$ is the unique subgroup of G of order $(\star)$.

- (b) Consider the following incomplete proof of (1) in the **Fundamental Theorem of Finite Cyclic Groups**:

The result is clear for $|H| = 1$. Otherwise, we know that H is cyclic, and so $H = \langle g^m \rangle$. Let $d = (A)$. Then $d|n$ and we only need to show that $H = \langle g^d \rangle$. From $d|m$, we have that $m = qd$ for some $q \in \mathbb{Z}$. Thus (B) and therefore $H \subseteq \langle g^d \rangle$. On the other hand, by (C) we have that $d = xm + yn$, where $x, y \in \mathbb{Z}$, and so

$$(D),$$

which implies that $\langle g^d \rangle \subseteq H$, and (1) follows.

With reference to this partial proof:

- i. 1 mark Complete the sentence at (A)
 - ii. 4 marks Fill in the necessary steps at (B) to conclude that $H \subseteq \langle g^d \rangle$.
 - iii. 1 mark Name the result you are using in (C).
 - iv. 4 marks Fill in the necessary steps at (D) to conclude that $\langle g^d \rangle \subseteq H$.
- (c) Consider the additive cyclic group $\mathbb{Z}_{18}$.
- i. 3 marks Determine and write down all the generators of $\mathbb{Z}_{18}$.
 - ii. 4 marks Find all the subgroups of $\mathbb{Z}_{18}$.
 - iii. 5 marks For each of the subgroups of $\mathbb{Z}_{18}$, write down all its elements.
 - iv. 1 mark Draw the lattice diagram of $\mathbb{Z}_{18}$.

Solution:

(a) FUNDAMENTAL THEOREM OF FINITE CYCLIC GROUPS. Let $G = \langle g \rangle$ be a cyclic group of order n .

- (1) If H is a subgroup of G , then $H = \langle g^d \rangle$ for some $d|n$.
- (2) If H is any subgroup of G with order $|H| = k$, then $k|n$.
- (3) If $k|n$, then $\langle g^{n/k} \rangle$ is the unique subgroup of G of order k .

(b) i. $d = \gcd(m, n)$.

ii. $g^m = g^{qd} = (g^d)^q \in \langle g^d \rangle$.

iii. Bézout's Lemma.

iv. $g^d = g^{xm+yn} = (g^m)^x(g^n)^y = (g^m)^x \in H$.

(c) Let us begin by recalling that

$$\mathbb{Z}_{18} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}, \bar{8}, \bar{9}, \bar{10}, \bar{11}, \bar{12}, \bar{13}, \bar{14}, \bar{15}, \bar{16}, \bar{17}\}$$

is an additive cyclic group generated by $\bar{1}$.

i. We know that an element $\bar{k} \in \mathbb{Z}_{18}$ is a generator of $\mathbb{Z}_{18}$ provided $\gcd(k, 18) = 1$. Using this we obtain that $\bar{1}, \bar{5}, \bar{7}, \bar{11}, \bar{13}, \bar{17}$ are all the generators of $\mathbb{Z}_{18}$.

ii and iii. To determine the subgroups of $\mathbb{Z}_{18}$, we use the Fundamental Theorem of Finite Cyclic Groups, which tells us that $\mathbb{Z}_{18}$ has exactly 6 subgroups of orders the positive divisor of 18, namely, 1, 2, 3, 6, 9, 18. The subgroups of orders 1 and 18 are clearly $\{\bar{0}\}$ and $\mathbb{Z}_{18}$, respectively.

For each $k \in \{2, 3, 6, 9\}$ the Fundamental Theorem of Finite Cyclic Groups tells us that $\langle \frac{18}{k}\bar{1} \rangle$ is the unique subgroup of $\mathbb{Z}_{18}$ of order k . Using this we obtain that

$$\text{order 2: } \left\langle \frac{18}{2}\bar{1} \right\rangle = \langle \bar{9} \rangle = \{\bar{0}, \bar{9}\},$$

$$\text{order 3: } \left\langle \frac{18}{3}\bar{1} \right\rangle = \langle \bar{6} \rangle = \{\bar{0}, \bar{6}, \bar{12}\},$$

$$\text{order 6: } \left\langle \frac{18}{6}\bar{1} \right\rangle = \langle \bar{3} \rangle = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}, \bar{12}, \bar{15}\},$$

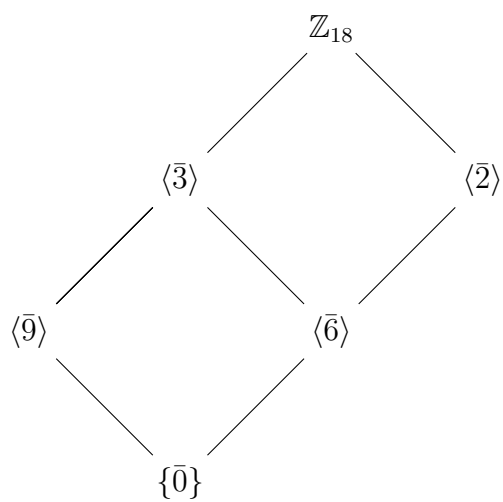
$$\text{order 9: } \left\langle \frac{18}{9}\bar{1} \right\rangle = \langle \bar{2} \rangle = \{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{10}, \bar{12}, \bar{14}, \bar{16}\}.$$

iv. Notice that

$$\langle \bar{9} \rangle \subseteq \langle \bar{3} \rangle,$$

$$\langle \bar{6} \rangle \subseteq \langle \bar{3} \rangle \quad \text{and} \quad \langle \bar{6} \rangle \subseteq \langle \bar{2} \rangle,$$

Using these we get the following lattice diagram:



4. (a) 4 marks State Bézout's Lemma.
- (b) 5 marks State the definition of a normal subgroup H of a group G ; give an example of a non-trivial normal subgroup of a group G (by non-trivial I mean that $H \neq \{1\}$ and $H \neq G$).
- (c) 5 marks State Lagrange's Theorem.
- (d) 10 marks Let K be a normal subgroup of a group G with index m in G ; that is, $|G : K| = m$. Suppose that n is a positive integer such that $\gcd(m, n) = 1$. Prove that K contains every element of G of order n .

Solution: (a) BÉZOUT'S LEMMA: Let m and n be two integers, not both zero. Then their $d := \gcd(n, m)$ exists and $d = xn + ym$ for some integers x and y .

(b) DEFINITION OF A NORMAL SUBGROUP: A subgroup H of a group G is called normal if $Hg = gH$ for all $g \in G$.

The centre $Z(G)$ is always a normal subgroup. Or any subgroup of an abelian group.

(c) LAGRANGE'S THEOREM: Let G be a finite group. Then the order of any subgroup of G divides the order of G .

(d) From $\gcd(n, m) = 1$, an application of Bézout's Lemma we get that $1 = xn + ym$ for some integers x and y . Suppose that g is an element of G order n , and let us prove that $g \in K$. To do so, we will use the coset properties (that we proved in class) and show that $Kg = K$.

Since the order of g is n we have that $g^n = 1$ and so

$$(Kg)^n = Kg^n = K \tag{1}$$

On the other hand, since $|G : K| = m$ and K is normal in G , we obtain that the factor group G/K has order m . An application of Lagrange's Theorem tells us that the order of the element Kg (which is, the order of the subgroup $\langle Kg \rangle$) divides m . In particular:

$$K = (Kg)^m = Kg^m \tag{2}$$

From (1) and (2) we get that

$$\begin{aligned} Kg &= Kg^{xn+ym} = (Kg)^{xn+ym} = (Kg)^{xn} \cdot (Kg)^{ym} = K(g^n)^x \cdot K(g^m)^y \\ &= K(1^x) \cdot K(1^y) = K \cdot K = K, \end{aligned}$$

as desired.